数理统计导论

Hogg, McKean, Craig

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Part 1

概率论基础

引言：从随机性中学习

1. 为什么需要概率论?

当我们掷一枚硬币时，结果是”正面”还是”反面?¨ 在单次实验中，答案似乎是确定的——但我们无法准确预测。这看似矛盾的现象揭示了一个深刻的真理：**随机性并非无知，而是世界的固有属性。**

数理统计（mathematical statistics）的核心问题是：*给定一组带有随机性的观测数据，我们如何从中提取关于未知真理的信息?*

要回答这个问题，我们必须首先拥有描述随机现象的数学语言。这正是概率论（probability theory）存在的意义。概率论提供了一套严谨的公理体系，使我们能够：

- 量化”可能性的¨ 大小（概率）；
- 描述随机变量的行为（分布）；
- 研究多个随机变量之间的关系（相关性、独立性）；
- 在大样本极限下揭示隐藏的规律（中心极限定理）。

正如微积分为物理学提供了描述连续变化的工具，概率论为数理统计提供了描述随机性的工具。**没有坚实的概率论基础，统计推断将成为无本之木。**

Remark. **这部分类比是否恰当? 微积分 *vs* 概率论**

2. 教材内容预览

本书（Hogg, McKean & Craig, *Introduction to Mathematical Statistics*）的前三章构建了概率论的基础设施，为后续的统计推断做好铺垫。这是一场从具体到抽象、从低维到高维的旅程。

2.1. 第一章：概率与分布（Probability and Distributions). 第一章从最基本的问题出发：*如何用数学语言描述”可能性?¨*

1. 概率公理化. 我们从集合论开始，建立样本空间（sample space）和事件（event）的概念。然后引入概率集函数（probability set function）P，这是一个满足非负性、归一化、可列可加性的集合函数。*为什么需要公理化? 因为我们需要一套在任何情况下都适用的严格规则，而不是依赖直觉。*

2. 分布函数. 随机变量（random variable）是将随机现象数量化的桥梁。对于随机变量 X，其累积分布函数（cumulative distribution function, CDF）定义为

$$F_X(x) = \mathbb{P}(X \leq x).$$

CDF 完整描述了随机变量的概率行为。对于离散型随机变量，我们用概率质量函数（probability mass function, pmf）$p_X(x) = \mathbb{P}(X = x)$；对于连续型随机变量，我们用概率密度函数（probability density function, pdf）$f_X(x)$，满足 $\mathbb{P}(a < X \leq b) = \int_a^b f_X(x)\,dx$。

为什么需要同时引入 *CDF*、*pmf*、*pdf*？这三种描述适用于不同场景。*CDF* 是最一般的概念；*pmf* 适用于离散型变量（如二项分布）；*pdf* 适用于连续型变量（如正态分布）。

3. 条件概率与贝叶斯公式. 条件概率（conditional probability）$P(A \mid B)$ 刻画了在已知 B 发生的前提下 A 发生的可能性。贝叶斯公式（Bayes' theorem）

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}$$

是统计推断的基石——它允许我们从"结果反向推断"原因的概率。

4. 条件期望. 条件期望（conditional expectation）$\mathbb{E}(X \mid Y = y)$ 是期望的概念在知道另一个变量信息下的推广。这在预测问题中至关重要：当我们知道 Y 的值时，X 的最佳预测恰好是条件期望。

5. 变换（Transformations）. 如果 X 是一个随机变量，当我们对其进行变换 $Y = g(X)$ 时，如何求 Y 的分布？这是统计学中的核心操作——我们经常需要对数据进行函数变换以简化分析。

6. 重要不等式. 本章以马尔可夫不等式（Markov inequality）和切比雪夫不等式（Chebyshev inequality）结尾。这些不等式建立了概率与矩之间的关系，尽管比较粗糙，但它们是证明大数定律等深刻结果的工具。

- **马尔可夫不等式:** 若 $X \geq 0$，则 $\mathbb{P}(X \geq t) \leq \mathbb{E}(X)/t$
- **切比雪夫不等式:** $\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \mathrm{var}(X)/t^2$

为什么关注不等式而不是精确概率？因为在信息不完整时，不等式提供了我们能获得的最好估计。

本章还包含 R 语言的基本使用——通过 Monte Carlo 模拟来验证理论结果。这体现了统计学的实践性：理论必须经受计算的检验。

2.2. 第二章：多元分布（Multivariate Distributions）. 第二章将我们从单变量世界带入多变量世界。为什么要研究多个随机变量？因为现实中的现象往往是多因素共同作用的结果——人的身高与体重、教育的投入与产出、药物的剂量与疗效。

1. 联合分布与边缘分布. 对于两个随机变量 X 和 Y，其联合分布（joint distribution）$F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y)$ 完整描述了它们的共同行为。从联合分布，我们可以"消去一个变量得到边缘分布（marginal distribution）：

$$F_X(x) = \lim_{y \to \infty} F_{XY}(x, y).$$

为什么需要边缘分布？因为我们有时只关心其中一个变量，而忽略另一个的影响。

2. 条件分布与条件期望. 给定 $Y = y$ 下 X 的条件分布为

$$F_{X|Y}(x \mid y) = \mathbb{P}(X \leq x \mid Y = y).$$

条件期望 $\mathbb{E}(X \mid Y = y)$ 是最佳预测——在均方误差意义下，它最小化了 X 的预测误差。

3. 相关系数. 相关系数（correlation coefficient）

$$\mathrm{corr}(X, Y) = \frac{\mathrm{cov}(X, Y)}{\sqrt{\mathrm{var}(X)\mathrm{var}(Y)}}$$

度量了两个变量之间的线性关系强度。为什么关注"线性关"系？因为线性相关是最容易处理的依赖形式，它连接了概率论与线性代数。

4. 矩阵形式与线性变换. 当我们推广到 n 维随机向量 $\mathbf{X} = (X_1, \ldots, X_n)'$ 时，协方差矩阵（covariance matrix）

$$\mathbf{\Sigma} = \mathrm{cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})']$$

成为了描述多变量随机性的核心工具。线性变换 $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ 的协方差矩阵为 $\mathbf{A}\mathbf{\Sigma}\mathbf{A}'$。

为什么使用矩阵语言？因为矩阵提供了处理高维数据的紧凑 *notation*，同时揭示了线性变换与二次型之间的深刻联系。

5. 变换. 与单变量情况类似，我们需要研究多变量变换：若 (X, Y) 的联合分布已知，如何求 $(U, V) = g(X, Y)$ 的联合分布？在 g 是双射时，我们有变量变换公式。

2.3. 第三章：一些特殊分布（Some Special Distributions). 第三章介绍统计学中最常用的概率分布。为什么要专门研究这些分布？因为它们在理论和应用中反复出现——掌握它们就像掌握基本词汇表。

1. 二项分布与相关分布. 二项分布（binomial distribution）$B(n, p)$ 描述了 n 次独立 Bernoulli 试验中成功次数的分布。其近亲包括：

- **几何分布:** 首次成功出现的试验次数
- **负二项分布:** 第 r 次成功出现的试验次数
- **超几何分布:** 不放回抽样时的成功次数（用于总体规模有限的情形）

2. 泊松分布. 泊松分布（Poisson distribution）$P(\lambda)$ 是稀有事件发生次数的极限分布。它具有独特的性质：若 $X \sim P(\lambda_1)$，$Y \sim P(\lambda_2)$ 且独立，则 $X + Y \sim P(\lambda_1 + \lambda_2)$（可加性）。

3. 伽马分布与卡方分布. 伽马分布（gamma distribution）$\Gamma(\alpha, \beta)$ 是连接多种分布的统一框架：

- 当 $\alpha = n/2$，$\beta = 1/2$ 时，得到**卡方分布** $\chi^2(n)$——在方差估计和拟合优度检验中核心地位
- 当 $\alpha = 1$ 时，得到指数分布
- 卡方分布的加法性质：若 $X_1 \sim \chi^2(n_1)$，$X_2 \sim \chi^2(n_2)$ 独立，则 $X_1 + X_2 \sim \chi^2(n_1 + n_2)$

为什么卡方分布如此重要？因为它出现在样本方差的抽样分布中——而样本方差是统计推断的核心量。

4. 贝塔分布. 贝塔分布（beta distribution）Beta(α, β) 是定义在 $(0, 1)$ 区间上的连续分布，常用于建模概率本身（当我们对概率参数 p 进行贝叶斯推断时，其后验分布往往是贝塔分布）。

5. 正态分布. 正态分布（normal distribution）$N(\mu, \sigma^2)$ 是统计学中最重要的分布。高斯（Gauss）发现测量误差服从正态分布；中心极限定理表明大量独立小因素的总和近似正态分布。其特殊地位体现在：

- 样本均值 $\bar{X}$ 的精确分布（若总体正态）或渐近分布（CLT）
- t 分布、F 分布的构造基础
- 多元正态分布是线性模型的理论支柱

6. t 分布与 F 分布. 这两个分布起源于小样本推断。当我们用样本标准差 S 替代总体标准差 σ 时，统计量

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

服从学生 t 分布（Student's t distribution）。在方差分析和回归分析中，F 分布用于比较模型拟合优度。

3. 这三章的内在逻辑

纵观前三章，我们可以看到一条清晰的主线：

(1) **描述单个随机变量.** 概率函数 → CDF/PDF → 期望与矩 → 特殊分布（第一、三章）

(2) **描述多个随机变量的关系.** 联合分布 → 边缘/条件分布 → 相关系数 → 线性变换（第二、七章）

(3) **连接理论与计算.** 理论结果 ↔ R 模拟验证（每一章都有）

这条主线将在后续章节中继续延伸：当我们研究估计量的大样本性质时（第五章），概率论的极限定理会再次登场；当我们构建最优检验时（第八章），充分性（第七章）的概念将揭示降维的奥秘。

统计学不是孤立的技巧集合，而是一个有机联系的理论体系。掌握这个体系的内在逻辑，比记住具体公式更为重要。

4. 学习方法建议

基于前三章的内容特点，我们建议：

- **重视定义.** 统计学的每一个概念——随机变量、分布函数、独立性、条件期望——都需要精确理解其定义。
- **多做计算.** 通过具体例子（如正态分布的标准化、二项分布的概率计算）来巩固抽象概念。

- **理解证明.** 关键定理（如全概率公式、贝叶斯公式、变量变换公式）的证明揭示了概念的深层联系。
- **结合 R 实践.** Monte Carlo 模拟帮助我们将抽象概率与具体数值联系起来。
- **建立联系.** 时刻问自己：这个概念与之前学过的内容有什么关联?

5. 习题说明

每章末尾的习题（Exercises）是学习的重要组成部分。习题类型包括：

- 概念验证题：检验对基本定义和定理的理解
- 计算题：练习具体概率和期望的计算
- 证明题：训练数学推导能力
- R 编程题：通过模拟验证理论结果

我们强烈建议在阅读正文后独立完成所有习题。统计学的理解只有在解决问题的过程中才能真正深化。

CHAPTER 1

Probability and Distributions

Chapter 1 概述

本章建立概率论的基本框架，从集合论和概率公理出发，逐步过渡到随机变量、期望、方差等核心概念。

1. 1.1 Introduction

本节直观介绍概率模型的概念，在 1.3 节中将其形式化。许多研究的特点是，在本质上相同的条件下可以重复进行实验。例如，医学研究关注药物的效果；经济学家关注某些商品的价格变化；农学家研究肥料对粮食产量的影响。实验产生 outcome，但无法在实验前确定预测结果。

若实验可以重复进行，且所有可能 outcome 可以事先描述，则称为随机实验（random experiment），所有可能 outcome 的集合称为样本空间（sample space），记为 $\mathcal{C}$。

Example 1.1 (例 1.1.1). *抛掷硬币，outcome 为 T（反面）或 H（正面）。样本空间为 $\mathcal{C} = \{H, T\}$。*

Example 1.2 (例 1.1.2). *投掷一枚红骰子和一枚白骰子，outcome 为有序对（红骰子点数，白骰子点数）。样本空间包含 36 个有序对：$\mathcal{C} = \{(1,1), \ldots, (1,6), (2,1), \ldots, (6,6)\}$。*

我们用小写 Roman 字母表示 $\mathcal{C}$ 的元素，如 a, b, c。事件（event）是 $\mathcal{C}$ 的子集，用大写 Roman 字母表示，如 A, B, C。

重复实验 N 次，事件 A 发生的次数（频率）记为 f，则 f/N 称为相对频率（relative frequency）。当 N 增大时，相对频率趋于稳定，趋向某个数 p，这称为事件 A 的概率（probability）。

Example 1.3 (例 1.1.3). *在例 1.1.2 中，设 B 为点数之和为 7 的所有有序对：$B = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$。若投掷 $N = 400$ 次，$f = 60$ 次出现点数之和为 7，则相对频率 $f/N = 60/400 = 0.15$，因此 B 的概率约为 0.15。*

Remark (注 1.1.1). *概率的相对频率解释依赖于实验可重复的条件。另一种观点是将概率视为主观信念。无论采用哪种解释，1.3 节的概率数学性质都是一致的。*

统计学的目的是为随机实验提供数学模型，进而进行统计推断。概率论的理论基础之一是基于集合和集合函数的概念，这正是下一节的内容。

2. 1.2 Sets

集合是现代数学的基本概念。设 C 表示区间 $[0,1]$ 上的实数集合，则 $\frac{3}{4} \in C$。

若集合 C 是有限的或与正整数集等势，则称 C 为可数的（countable）。

2.1. 1.2.1 Review of Set Theory. 设 $\mathcal{C}$ 为样本空间，事件是 $\mathcal{C}$ 的子集。

Definition 1.1 (定义 1.2.1). *事件 A 的补（complement）A^c 是 $\mathcal{C}$ 中所有不属于 A 的元素构成的集合：$A^c = \{x \in \mathcal{C} : x \notin A\}$。*

空集（empty set）记为 ϕ。注意 $\mathcal{C}^c = \phi$，$\phi^c = \mathcal{C}$。

Definition 1.2 (定义 1.2.2). *若集合 A 的每个元素都是集合 B 的元素，则称 A 是 B 的子集，记作 $A \subset B$。若 $A \subset B$ 且 $B \subset A$，则 $A = B$。*

Definition 1.3 (定义 1.2.3). *设 A 和 B 为事件。A 和 B 的并（union）$A \cup B$ 是所有属于 A 或属于 B 的元素构成的集合。*

Definition 1.4 (定义 1.2.4). *设 A 和 B 为事件。A 和 B 的交（intersection）$A \cap B$ 是所有同时属于 A 和 B 的元素构成的集合。*

Definition 1.5 (定义 1.2.5). *若 $A \cap B = \phi$，则称 A 和 B 是不交的（disjoint）。*

若 $A_1, A_2, \ldots, A_n$ 为任意集合，定义

$$A_1 \cup A_2 \cup \cdots \cup A_n = \{x : x \in A_i \text{ for some } i = 1, 2, \ldots, n\} \tag{1.2.8}$$

$$A_1 \cap A_2 \cap \cdots \cap A_n = \{x : x \in A_i \text{ for all } i = 1, 2, \ldots, n\}. \tag{1.2.9}$$

简写为 $\cup_{i=1}^n A_i$ 和 $\cap_{i=1}^n A_i$。

可数并和可数交定义为

$$\cup_{n=1}^\infty A_n = \{x : x \in A_n \text{ for some } n\} \tag{1.2.10}$$

$$\cap_{n=1}^\infty A_n = \{x : x \in A_n \text{ for all } n\}. \tag{1.2.11}$$

德摩根定律（DeMorgan's Laws）：对任意集合 A 和 B，

$$(A \cap B)^c = A^c \cup B^c \tag{1.2.6}$$

$$(A \cup B)^c = A^c \cap B^c. \tag{1.2.7}$$

分配律：

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \tag{1.2.5}$$

Example 1.4 (例 1.2.3). *设 $\mathcal{C} = \{1, 2, 3, \ldots\}$，$A_n = \{1, 3, \ldots, 2n-1\}$，$B_n = \{n, n+1, \ldots\}$，则*

$$\cup_{n=1}^\infty A_n = \{1, 3, 5, \ldots\}; \quad \cap_{n=1}^\infty A_n = \{1\}$$

$$\cup_{n=1}^\infty B_n = \mathcal{C}; \quad \cap_{n=1}^\infty B_n = \phi. \tag{1.2.13}$$

单调序列：若 $\{A_n\}$ 非递减（$A_n \subset A_{n+1}$），则 $\lim_{n\to\infty} A_n = \cup_{n=1}^\infty A_n$。若 $\{A_n\}$ 非递增（$A_n \supset A_{n+1}$），则 $\lim_{n\to\infty} A_n = \cap_{n=1}^\infty A_n$。

2.2. 1.2.2 Set Functions. 集合函数（set function）是将集合映射到实数的函数。

Example 1.5 (例 1.2.5). 设 $\mathcal{C}=\mathbb{R}$，$Q(A)$ 为 A 中正整数的个数。若 $A=\{x:0<x<5\}$，则 $Q(A)=4$；若 $A=\{-2,-1\}$，则 $Q(A)=0$。

Example 1.6 (例 1.2.6). 设 $\mathcal{C}=\mathbb{R}^2$，$Q(A)$ 为 A 的面积（若面积存在）。若 $A=\{(x,y):x^2+y^2\leq 1\}$（单位圆盘），则 $Q(A)=\pi$。

集合函数常用求和或积分定义。记号:

$$\sum_A f(x) \quad \text{表示在所有} x\in A \text{ 上求和}$$

$$\int_A f(x)\,dx \quad \text{表示在集合} A \text{ 上的积分}$$

Example 1.7 (例 1.2.7). 设 $\mathcal{C}$ 为所有非负整数构成的集合，$Q(A)=\sum_{n\in A}\left(\frac{2}{3}\right)^n$。由几何级数公式 $\sum_{n=0}^{\infty}a^n=\frac{1}{1-a}$（$|a|<1$），可得 $Q(\mathcal{C})=3$。

Example 1.8 (例 1.2.8). 设 $\mathcal{C}=(0,\infty)$，$Q(A)=\int_A e^{-x}\,dx$。则

$$Q[(1,3)]=\int_1^3 e^{-x}\,dx=e^{-1}-e^{-3}\approx 0.318$$

$$Q(\mathcal{C})=\int_0^\infty e^{-x}\,dx=1.$$

3. 1.3 The Probability Set Function

设 $\mathcal{C}$ 为样本空间，$\mathcal{B}$ 为事件域（σ-域），现在定义概率集函数。

Remark 1.3.1: 概率的定义由三条公理组成，这三条公理来源于相对频率的三条直观性质。设重复实验 N 次，事件 A 的相对频率为 $f_A=\#\{A\}/N$。则 $f_A\geq 0$、$f_{\mathcal{C}}=1$，且若 A_1,A_2 不交则 $f_{A_1\cup A_2}=f_{A_1}+f_{A_2}$。

Definition 1.6 (定义 1.3.1 (概率)). 设 $\mathcal{C}$ 为样本空间，$\mathcal{B}$ 为事件域。函数 $\mathbb{P}:\mathcal{B}\to\mathbb{R}$ 满足以下三条条件则称为概率集函数:

(1) $\mathbb{P}(A)\geq 0$，对所有 $A\in\mathcal{B}$

(2) $\mathbb{P}(\mathcal{C})=1$

(3) 若 $\{A_n\}$ 为两两不相交的事件序列，则

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty}A_n\right)=\sum_{n=1}^{\infty}\mathbb{P}(A_n).$$

若事件序列两两不相交，称其为互不相容的（mutually exclusive）。若序列还是穷尽的（exhaustive），即其并为 $\mathcal{C}$，则构成 $\mathcal{C}$ 的一个划分（partition）。

概率集函数告诉我们在事件域上概率如何分布。

Theorem 1.1 (定理 1.3.1). 对每个事件 $A\in\mathcal{B}$，$\mathbb{P}(A)=1-\mathbb{P}(A^c)$。

证明. 由 $\mathcal{C} = A \cup A^c$ 和 $A \cap A^c = \phi$，结合公理 (2) 和 (3) 得

$$1 = \mathbb{P}(A) + \mathbb{P}(A^c).$$

□

THEOREM 1.2 (定理 1.3.2). $\mathbb{P}(\phi) = 0$。

THEOREM 1.3 (定理 1.3.3). *若 $A \subset B$，则 $\mathbb{P}(A) \leq \mathbb{P}(B)$。*

证明. $B = A \cup (A^c \cap B)$，且两项不相交。由公理 (3) 得 $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B) \geq \mathbb{P}(A)$。 □

COROLLARY 1.1 (推论 1.3.1). *对每个 $A \in \mathcal{B}$，$0 \leq \mathbb{P}(A) \leq 1$。*

THEOREM 1.4 (定理 1.3.5). *对任意事件 A, B，*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

证明. 将 $A \cup B$ 和 B 分解为不相交集合的并：

$$A \cup B = A \cup (A^c \cap B), \quad B = (A \cap B) \cup (A^c \cap B).$$

由公理 (3) 得 $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B)$ 和 $\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B)$。消去 $\mathbb{P}(A^c \cap B)$ 即得结果。 □

等可能性情形（Equilikely Case）：若样本空间 $\mathcal{C} = \{x_1, x_2, \ldots, x_m\}$ 有限，且每个 outcome 等概率 $p_i = 1/m$，则

$$\mathbb{P}(A) = \sum_{x_i \in A} \frac{1}{m} = \frac{\#(A)}{m}.$$

4. 1.4 Conditional Probability and Independence

4.1. Conditional Probability.

DEFINITION 1.7 (定义 1.4.1 (条件概率)). *设 $\mathbb{P}(B) > 0$，在事件 B 已发生的条件下，事件 A 的条件概率（conditional probability）定义为*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

条件概率也满足概率公理。

由条件概率定义可直接导出乘法法则（multiplication rule）：

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B \mid A) = \mathbb{P}(B) \cdot \mathbb{P}(A \mid B).$$

推广到 n 个事件：

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}\left(A_n \mid \bigcap_{i=1}^{n-1} A_i\right).$$

全概率公式（Law of Total Probability）：设 $\{B_i\}$ 为 $\mathcal{C}$ 的一个划分，则对任意事件 A，

$$\mathbb{P}(A) = \sum_i \mathbb{P}(B_i) \cdot \mathbb{P}(A \mid B_i).$$

THEOREM 1.5 (定理 1.4.1 (贝叶斯公式)). 设 $\{B_i\}$ 为 $\mathcal{C}$ 的一个划分，$\mathbb{P}(B_i) > 0$。则对任意事件 A（$\mathbb{P}(A) > 0$），

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(B_j) \cdot \mathbb{P}(A \mid B_j)}{\sum_i \mathbb{P}(B_i) \cdot \mathbb{P}(A \mid B_i)}.$$

术语：$\mathbb{P}(B_j)$ 为先验概率（prior probability），$\mathbb{P}(B_j \mid A)$ 为后验概率（posterior probability），$\mathbb{P}(A \mid B_j)$ 为似然（likelihood）。

EXAMPLE 1.9 (例 1.4.1 (医学检测)). 设某疾病患病率为 1%，检测准确率为 99%。令 $D = \{\text{患病}\}$，$+ = \{\text{检测阳性}\}$。则

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(D) \cdot \mathbb{P}(+ \mid D)}{\mathbb{P}(D) \cdot \mathbb{P}(+ \mid D) + \mathbb{P}(D^c) \cdot \mathbb{P}(+ \mid D^c)} = \frac{0.01 \cdot 0.99}{0.01 \cdot 0.99 + 0.99 \cdot 0.01} = \frac{1}{2}.$$

即使检测准确率高达 99%，检测阳性者真正患病的概率只有 50%!

REMARK (贝叶斯视角的启示). 贝叶斯公式不仅仅是一个计算工具，更提供了一种**思考不确定性的思维方式**。

***1.* 先验与后验的博弈**

先验概率（*prior*）$\mathbb{P}(B_j)$ 反映了我们在获得新证据之前的信念；后验概率（*posterior*）$\mathbb{P}(B_j \mid A)$ 则是融合了新证据之后的更新信念。贝叶斯公式告诉我们：如何根据证据更新信念。

***2.* 似然的核心作用**

似然（*likelihood*）$\mathbb{P}(A \mid B_j)$ 衡量的是，在假设 B_j 成立的条件下，观测到证据 A 的"理所当然"程度。注意这是一个方向性问题——它不是问"B_j 能否解释 A"，而是问"如果 B_j 为真，A 有多常见"。这正是贝叶斯推理与逻辑推理的关键区别。

***3.* 从特殊到一般的延伸**

在医学检测例子中，我们关注的是 $\mathbb{P}(\text{患病} \mid \text{阳性})$——这是最直接有用的信息。但在更一般的统计推断中，我们往往需要从数据（A）反推参数（B_j）的分布。这正是贝叶斯统计的核心任务。

***4.* 与频率学派的对话**

频率学派将概率解释为长期频率，认为参数是固定常数，只能通过反复抽样来估计；贝叶斯学派则将参数也视为随机变量，用概率分布描述我们对参数的信念，并将数据视为更新这种信念的证据。两种视角并非互斥——贝叶斯公式本身是中性的，它只是提供了融合先验信息与观测数据的数学框架。

4.2. 1.4.1 Independence.

DEFINITION 1.8 (定义 1.4.2 (独立性)). 若 $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$，则称事件 A 与 B 相互独立（*independent*），记作 $A \perp\!\!\!\perp B$。

当 $\mathbb{P}(A) > 0, \mathbb{P}(B) > 0$ 时，独立性等价于 $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ 或 $\mathbb{P}(B \mid A) = \mathbb{P}(B)$。

Proposition 1.1 (命题 1.4.1). *若 $A \perp\!\!\!\perp B$，则：*

(1) $A \perp\!\!\!\perp B^c$

(2) $A^c \perp\!\!\!\perp B$

(3) $A^c \perp\!\!\!\perp B^c$

Definition 1.9 (定义 1.4.3 (相互独立)). *事件序列 $\{A_i : i \in I\}$ 相互独立，若对任意有限子集 $\{i_1, \ldots, i_k\}$，*

$$\mathbb{P}\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} \mathbb{P}(A_{i_j}).$$

Warning: 两两独立（pairwise independent）不一定意味着相互独立!

5. 1.5 Random Variables

随机变量（random variable）是从样本空间 $\mathcal{C}$ 到实数的函数：$X : \mathcal{C} \to \mathbb{R}$。

通常用大写字母 X, Y, Z 表示随机变量，用小写字母 x, y, z 表示具体取值。

随机变量可分为：

- **离散随机变量（discrete）**：取值可数
- **连续随机变量（continuous）**：取值充满区间，不可数

Example 1.10 (例 1.5.1). *抛掷两枚硬币。样本空间 $\mathcal{C} = \{HH, HT, TH, TT\}$。定义 X 为正面出现的次数：$X(HH) = 2$，$X(HT) = X(TH) = 1$，$X(TT) = 0$。则 X 是取值在 $\{0, 1, 2\}$ 的离散随机变量。*

6. 1.6 Discrete Random Variables

Definition 1.10 (定义 1.6.1 (概率质量函数)). *设 X 为离散随机变量。函数 $p_X(x) = \mathbb{P}(X = x)$ 称为概率质量函数（probability mass function, pmf）。*

pmf 满足：$p_X(x) \geq 0$ 对所有 x，且 $\sum_x p_X(x) = 1$。

伯努利分布（Bernoulli）：$X \sim \text{Bernoulli}(p)$，取值 $\{0, 1\}$，

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

二项分布（Binomial）：$X \sim \text{Bin}(n, p)$，

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

这是 n 次独立伯努利试验中成功的次数。

泊松分布（Poisson）：$X \sim \text{Poisson}(\lambda)$，

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots.$$

泊松分布是二项分布当 $n \to \infty$、$p \to 0$、$np = \lambda$ 固定时的极限分布。

6.1. 1.6.1 Transformations. 设 X 为离散随机变量，$Y = g(X)$。则 $p_Y(y) = \sum_{x:g(x)=y} p_X(x)$。

7. 1.7 Continuous Random Variables

Definition 1.11 (定义 1.7.1 (概率密度函数)). *设 X 为连续随机变量。若存在非负函数 $f_X(x)$ 使得*

$$\mathbb{P}(a < X < b) = \int_a^b f_X(x)\,dx$$

对任意 $a < b$ 成立，则称 $f_X(x)$ 为概率密度函数（probability density function, pdf）。

pdf 满足：$f_X(x) \geq 0$ 对所有 x，且 $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$。
Warning: $f_X(x)$ 可以大于 1!
正态分布（Normal/Gaussian）：$X \sim N(\mu, \sigma^2)$，

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

标准正态分布：$Z \sim N(0,1)$，$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$。
指数分布（Exponential）：$X \sim \mathrm{Exp}(\lambda)$，$f(x) = \lambda e^{-\lambda x}$，$x > 0$。

7.1. 1.7.1 Quantiles.

Definition 1.12 (定义 1.7.2 (分位数)). *设 X 的 CDF 为 $F_X(x)$。对 $0 < p < 1$，p-分位数 x_p 满足 $F_X(x_p) = p$。中位数是 $p = 0.5$ 的分位数。*

7.2. 1.7.2 Transformations. 设 X 为连续随机变量，$Y = g(X)$。若 $y = g(x)$ 严格单调且可导，则

$$f_Y(y) = f_X(h(y)) \cdot |h'(y)|,$$

其中 $x = h(y)$ 是 $y = g(x)$ 的反函数。

若 $Y = aX + b$，则 $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$。

8. 1.8 Expectation of a Random Variable

Definition 1.13 (定义 1.8.1 (数学期望)). *随机变量 X 的期望（expected value）：*

- *离散型：$\mathbb{E}X = \sum_x x \cdot p_X(x)$*
- *连续型：$\mathbb{E}X = \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx$*

要求级数或积分绝对收敛。

Theorem 1.6 (定理 1.8.1 (期望的线性性)). *对任意随机变量 X, Y 和常数 a, b:*

$$\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y.$$

关键点：*无需 X, Y 相互独立!*

THEOREM 1.7 (定理 1.8.2 (方差公式)).

$$var(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2,$$

其中 $\mu = \mathbb{E}X$。

THEOREM 1.8 (定理 1.8.3 (方差的性质)). *(1)* $var(aX+b) = a^2 var(X)$
(2) 若 $X_1, \ldots, X_n$ 相互独立，则 $var(\sum_{i=1}^n X_i) = \sum_{i=1}^n var(X_i)$

9. 1.9 Some Special Expectations

DEFINITION 1.14 (定义 1.9.1 (协方差)).

$$cov(X,Y) = \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)] = \mathbb{E}(XY) - \mathbb{E}X \cdot \mathbb{E}Y.$$

协方差的性质：

(1) $\mathrm{cov}(X,X) = \mathrm{var}(X)$
(2) $\mathrm{cov}(X,Y) = \mathrm{cov}(Y,X)$
(3) $\mathrm{cov}(aX+b,Y) = a\mathrm{cov}(X,Y)$
(4) $\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\mathrm{cov}(X,Y)$
(5) 若 $X \perp\!\!\!\perp Y$，则 $\mathrm{cov}(X,Y) = 0$

DEFINITION 1.15 (定义 1.9.2 (相关系数)).

$$corr(X,Y) = \frac{cov(X,Y)}{\sqrt{var(X) \cdot var(Y)}}.$$

$-1 \le \mathrm{corr}(X,Y) \le 1$。

10. 1.10 Important Inequalities

THEOREM 1.9 (Markov 不等式). 若 X 为随机变量，$g(X) \ge 0$，则对任意 $a > 0$，

$$\mathbb{P}(g(X) \ge a) \le \frac{\mathbb{E}[g(X)]}{a}.$$

特别地，取 $g(X) = X^2$，$a = k^2$，得

$$\mathbb{P}(|X| \ge k) \le \frac{\mathbb{E}[X^2]}{k^2}.$$

THEOREM 1.10 (Chebyshev 不等式). 设 $\mu = \mathbb{E}X$，$\sigma^2 = var(X)$。则对任意 $k > 0$，

$$\mathbb{P}(|X-\mu| \ge k) \le \frac{\sigma^2}{k^2}.$$

THEOREM 1.11 (Cauchy-Schwarz 不等式).

$$|\mathbb{E}(XY)| \le \sqrt{\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]}.$$

Summary

本章建立了概率论的基本框架：

(1) **概率公理化**：从相对频率的直观性质出发，建立了概率的公理化定义
(2) **条件概率与贝叶斯公式**：提供了更新信息的数学工具
(3) **独立性**：刻画了两个事件之间的概率关系
(4) **随机变量**：从集合论过渡到数值函数，为分布理论奠定基础
(5) **期望与方差**：分别度量分布的中心位置和离散程度
(6) **协方差与相关系数**：度量两个随机变量的联合变化趋势
(7) **重要不等式**：Markov、Chebyshev、Cauchy-Schwarz

Exercises

Exercise 1.1. *1.2.1* 求 $C_1 \cup C_2$ 和 $C_1 \cap C_2$：

(1) $C_1 = \{0, 1, 2\}$，$C_2 = \{2, 3, 4\}$
(2) $C_1 = \{x : 0 < x < 2\}$，$C_2 = \{x : 1 \leq x < 3\}$

Exercise 1.2. *1.3.1* 设 $\mathbb{P}(A) = 0.3$，$\mathbb{P}(B) = 0.4$，$\mathbb{P}(A \cap B) = 0.1$。求：

(1) $\mathbb{P}(A \cup B)$
(2) $\mathbb{P}(A \mid B)$
(3) $\mathbb{P}(B \mid A)$

Exercise 1.3. *1.4.1* 某箱中有 *5* 个红球和 *3* 个白球。无放回地抽取 *3* 个球。求恰好抽到 *2* 个红球的概率。

Exercise 1.4. *1.5.1* 设 $X \sim N(100, 225)$。求：

(1) $\mathbb{P}(X < 95)$
(2) $\mathbb{P}(90 < X < 110)$

CHAPTER 2

二元随机变量分布

Chapter 2 概述

在实际问题中，我们往往不只关心单个随机变量，而是关心它们之间的**相互作用**。例如，在金融中我们想研究两种资产收益的联合波动；在医学中想了解剂量与疗效的关系；在物理中想描述两个测量值的相关性。本章将一元推广到二元，建立联合分布理论。

本章路线图：联合分布 → 边缘分布 → 条件分布 → 独立性 → 相关性 → 二元正态

1. 2.1 从一元到二元

1.1. 为什么需要联合分布?. 在一元情形，我们用 cdf $F_X(x) = \mathbb{P}(X \le x)$ 描述单个随机变量的行为。但在许多场景中，*单个变量的分布并不能回答我们的问题*——我们需要知道多个变量如何共同变化。这引导我们提出以下问题：

核心问题：给定两个随机变量 X_1 和 X_2，如何描述它们的联合行为?

这促使我们定义随机向量和联合分布。

1.2. 随机向量.

Definition 2.1 (随机向量). *设样本空间为 $\mathcal{C}$，X_1 和 X_2 为定义在 $\mathcal{C}$ 上的两个随机变量。则 (X_1, X_2) 称为随机向量，其取值空间为 $\mathcal{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in \mathcal{C}\}$。*

为什么要用向量? 因为 (X_1, X_2) 的取值是一个有序对，用向量符号 $\mathbf{X} = (X_1, X_2)'$ 能更清晰地表示这种结构。

1.3. 联合分布. 联合 cdf 是一元 cdf 的自然推广：

$$F_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 \le x_1, X_2 \le x_2). \tag{1}$$

离散型：用 pmf 描述，

$$p_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2), \tag{2}$$

满足 $p \ge 0$ 且 $\sum\sum_{\mathcal{D}} p = 1$。

连续型：用 pdf 描述，

$$F_{X_1,X_2}(x_1, x_2) = \int_{-\infty}^{x_1}\int_{-\infty}^{x_2} f(w_1, w_2)\, dw_1\, dw_2, \tag{3}$$

其中 $f \ge 0$ 且 $\int\int_{\mathcal{D}} f = 1$。

Warning：与一元情形不同，这里的 $f(x_1, x_2)$ 可以大于 1！因为它是”密度”而非概率值。

Remark. 几何直观 若把联合 *pdf* $z = f(x_1, x_2)$ 视作一张曲面，则 $\mathbb{P}((X_1, X_2) \in A)$ 就是该曲面下区域 A 上的体积。这是一元情形"曲线下面积"在二维的推广。emark

1.4. 从联合到边缘：积分即求和. 核心问题：已知 (X_1, X_2) 的联合分布，如何得到单个 X_1 的分布?

直觉上，我们需要"忽略" X_2 的信息。具体做法是：

- **离散型**：对 x_2 求和
- **连续型**：对 x_2 积分

这引导我们得到：

边缘 pmf：

$$p_{X_1}(x_1) = \sum_{x_2} p_{X_1,X_2}(x_1, x_2). \tag{4}$$

边缘 pdf：

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)\, dx_2. \tag{5}$$

Remark. 为什么叫"边缘"? 在二维离散情形，将联合 *pmf* 列表，边缘分布恰好是每行（或每列）的"边缘"之和。这解释了名称的由来。emark

1.5. 期望的推广. 同样的思想：$Y = g(X_1, X_2)$ 是新的随机变量，其期望为

$$\mathbb{E}Y = \int\int g(x_1, x_2) f(x_1, x_2)\, dx_1\, dx_2 \quad \text{（连续型）}.$$

关键发现：期望算子 E 对加法保持线性，无论 X_1, X_2 是否独立：

$$\mathbb{E}(aY_1 + bY_2) = a\mathbb{E}Y_1 + b\mathbb{E}Y_2. \tag{6}$$

这为什么重要? 在第 1 章我们学过独立随机变量之和的方差公式 $\mathrm{var}(X_1 + X_2) = \mathrm{var}(X_1) + \mathrm{var}(X_2)$。期望的线性性是其基础——它无需独立性即可成立。

2. 2.2 变换：寻找新视角

给定 (X_1, X_2) 的分布，我们经常想研究其函数 $Y = u(X_1, X_2)$ 的分布。例如：

- 和 $Y = X_1 + X_2$（两个随机变量之和）
- 商 $Y = X_1/X_2$（比值）
- 极坐标变换 $(R, \Theta) = (\sqrt{X_1^2 + X_2^2}, \arctan(X_2/X_1))$

对于连续型，一一对应的变换有简洁的公式：

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}(v_1(y_1, y_2), v_2(y_1, y_2)) \cdot |J|. \tag{7}$$

其中 Jacobian 行列式

$$J = \det \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{pmatrix} = \frac{\partial x_1}{\partial y_1} \cdot \frac{\partial x_2}{\partial y_2} - \frac{\partial x_1}{\partial y_2} \cdot \frac{\partial x_2}{\partial y_1}. \tag{8}$$

为什么有 $|J|$?这是面积元素的变换因子。在一元情形我们有 $f_Y(y) = f_X(h(y))|h'(y)|$；二维情形类似，只是用行列式度量了面积缩放比例。

3. 2.3 条件分布：信息的价值

核心问题：已知 X_1 的值，X_2 的分布会如何变化?

这就是条件分布要回答的问题。条件分布本质上是在 $X_1 = x_1$ 的条件下，重新审视 X_2 的行为。

离散型条件 pmf：

$$p_{X_2|X_1}(x_2 \mid x_1) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_1}(x_1)}, \quad p_{X_1}(x_1) > 0. \tag{9}$$

连续型条件 pdf：

$$f_{X_2|X_1}(x_2 \mid x_1) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_1}(x_1)}, \quad f_{X_1}(x_1) > 0. \tag{10}$$

注意：分子是联合密度，分母是边缘密度。这个比值衡量的是，在 $X_1 = x_1$ 的条件下，(x_1, x_2) 这个”点”有多”典型”。

由此可以定义条件期望：

$$\mathbb{E}[u(X_2) \mid x_1] = \int_{-\infty}^{\infty} u(x_2) f_{X_2|X_1}(x_2 \mid x_1)\, dx_2. \tag{11}$$

这里有一个深刻的定理，它连接了条件与边缘：

$$\mathbb{E}[\mathbb{E}(X_2 \mid X_1)] = \mathbb{E}(X_2). \tag{12}$$

这称为**全期望公式**。它的含义是：*对条件均值再取期望，等于无条件期望*。直观上，无论我们按哪个 x_1 分组计算均值，最终的平均仍然是总均值。

更惊人的是方差的不等式：

$$\mathrm{var}[\mathbb{E}(X_2 \mid X_1)] \le \mathrm{var}(X_2).$$

Rao-Blackwell 定理：若用 X_2 估计 $\mu_2 = \mathbb{E}X_2$，则条件均值 $E(X_2 \mid x_1)$ 比 X_2 本身更精确（方差更小）。这在统计推断中极其重要——它告诉我们如何利用辅助信息改进估计。

4. 2.4 独立性：变量间的”无关系”

核心问题：什么时候 X_1 和 X_2 互不影响?

若 X_1 与 X_2 独立，则知道 X_1 的值不会改变我们对 X_2 的认知。用条件分布的语言：

$$f_{X_2|X_1}(x_2 \mid x_1) = f_{X_2}(x_2) \quad （不依赖于x_1）.$$

由此推出联合密度可分离：

$$X_1 \text{ 与} X_2 \text{ 独立} \iff f(x_1, x_2) \equiv f_1(x_1) f_2(x_2). \tag{13}$$

一个实用的判据：

$$X_1 \text{ 与} X_2 \text{ 独立} \iff f(x_1, x_2) \equiv g(x_1)h(x_2) \quad \text{（可分离变量）}. \tag{14}$$

独立性的后果（这些在第 1 章学过，现在有了二元语境）：

- $\mathbb{E}[g(X_1)h(X_2)] = \mathbb{E}[g(X_1)] \cdot \mathbb{E}[h(X_2)]$
- $\mathrm{cov}(X_1, X_2) = 0$

Remark. *重要警告* **逆命题不成立！** *$cov(X_1, X_2) = 0$ 并不意味着独立性（除二元正态分布外）。*emark

这提醒我们：**不相关**（uncorrelated）和**独立**（independent）是两个不同概念。不相关只说明没有线性关系，但变量间可能存在非线性依赖。

5. 2.5 相关性：线性关联的度量

我们已经知道 $\mathrm{cov}(X_1, X_2) = 0$ 不能完全刻画独立性。那么如何量化 X_1 与 X_2 之间的" 关联程度"?

协方差 $\mathrm{cov}(X_1, X_2)$ 存在一个问题：它依赖于 X_1, X_2 的量纲。例如，身高（cm）和体重（kg）的协方差与换算成英寸和磅时不同。

解决方案：先标准化，再求协方差：

$$\rho_{12} = \mathrm{corr}(X_1, X_2) = \frac{\mathrm{cov}(X_1, X_2)}{\sqrt{\mathrm{var}(X_1)}\sqrt{\mathrm{var}(X_2)}} = \frac{\mathrm{cov}(X_1, X_2)}{\sigma_1 \sigma_2}. \tag{15}$$

这就是相关系数（correlation coefficient）。

核心性质：

- $-1 \le \rho_{12} \le 1$
- $|\rho_{12}| = 1 \iff X_2 = aX_1 + b$（完美线性关系）
- $\rho_{12} = 0$ 称**不相关**

为什么是 $[-1, 1]$**?** 这来自 Cauchy-Schwarz 不等式：

$$|\mathrm{cov}(X_1, X_2)| = |\mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)]| \le \sqrt{\mathbb{E}[(X_1 - \mu_1)^2] \cdot \mathbb{E}[(X_2 - \mu_2)^2]} = \sigma_1 \sigma_2.$$

Remark. *相关 vs 协方差 协方差取决于量纲（" 米的协方差"vs" 厘米的协方差" 数值不同）；相关系数是无量纲的，消除了单位影响。因此，相关系数是更通用的" 关联程度" 度量。*emark

6. 2.6 二元正态：最重要的二元分布

在所有二元分布中，二元正态分布占据核心地位。它是一元正态分布在二维的自然推广。

为什么重要?

(1) 边缘分布和条件分布都是正态的

(2) 线性组合仍是正态

(3) 不相关 $\iff$ 独立（这是正态独有的优雅性质）
(4) 多元正态是计量经济学、信号处理等领域的基石

Definition 2.2 (二元正态分布). (X_1, X_2) 服从二元正态分布，若其联合 *pdf* 为

$$f(x_1,x_2)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2}-2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2}+\frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right\},$$

记作 $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$。

五个参数的含义:

- μ_1, μ_2：边缘分布的均值
- σ_1^2, σ_2^2：边缘分布的方差
- ρ：相关系数

Remark. 条件分布的几何直觉 在二元正态中，条件均值

$$\mathbb{E}(X_2 \mid x_1)=\mu_2+\rho\frac{\sigma_2}{\sigma_1}(x_1-\mu_1)$$

是 x_1 的线性函数。这意味着：给定 $X_1 = x_1$，X_2 的最佳预测也是线性的，其斜率由相关系数 ρ 决定。emark

关键性质:

(1) 边缘分布：$X_1 \sim N(\mu_1, \sigma_1^2)$，$X_2 \sim N(\mu_2, \sigma_2^2)$
(2) 条件分布：

$$X_2 \mid X_1 = x_1 \sim N\left(\mu_2+\rho\frac{\sigma_2}{\sigma_1}(x_1-\mu_1),\ \sigma_2^2(1-\rho^2)\right)$$

(3) $X_1 \perp\!\!\!\perp X_2 \iff \rho = 0$（正态独有的" 不相关 $\Rightarrow$ 独立"）
(4) 任意线性组合 $aX_1 + bX_2 \sim N(a\mu_1 + b\mu_2, \mathrm{var}(aX_1 + bX_2))$

概念	公式/描述
联合 cdf	$F_{X_1,X_2}(x_1,x_2)=\mathbb{P}(X_1\le x_1, X_2\le x_2)$
边缘 pmf	$p_{X_1}(x_1)=\sum_{x_2}p_{X_1,X_2}(x_1,x_2)$
边缘 pdf	$f_{X_1}(x_1)=\int_{-\infty}^{\infty}f(x_1,x_2)\,dx_2$
条件 pmf	$p_{X_2\mid X_1}(x_2\mid x_1)=p(x_1,x_2)/p_{X_1}(x_1)$
条件 pdf	$f_{X_2\mid X_1}(x_2\mid x_1)=f(x_1,x_2)/f_{X_1}(x_1)$
全期望	$\mathbb{E}[\mathbb{E}(X_2\mid X_1)]=\mathbb{E}X_2$
独立性	$f(x_1,x_2)=f_1(x_1)f_2(x_2)$
协方差	$\mathrm{cov}=\mathbb{E}(X_1X_2)-\mathbb{E}X_1\mathbb{E}X_2$
相关系数	$\rho=\mathrm{cov}/(\sigma_1\sigma_2)$，$-1\le\rho\le 1$

CHAPTER 3

Some Special Distributions

1. The Binomial and Related Distributions

In Chapter 1 we introduced the uniform distribution and the hypergeometric distribution. In this chapter we discuss some other important distributions of random variables frequently used in statistics. We begin with the binomial and related distributions.

A Bernoulli experiment is a random experiment, the outcome of which can be classified in but one of two mutually exclusive and exhaustive ways, for instance, success or failure (e.g., female or male, life or death, nondefective or defective). A sequence of Bernoulli trials occurs when a Bernoulli experiment is performed several independent times so that the probability of success, say p , remains the same from trial to trial.

Let X be a random variable associated with a Bernoulli trial by defining it as follows:

$$X(\text{success}) = 1 \quad \text{and} \quad X(\text{failure}) = 0.$$

That is, the two outcomes, success and failure, are denoted by one and zero, respectively. The pmf of X can be written as

$$p(x) = p^x(1-p)^{1-x}, \quad x = 0, 1, \tag{16}$$

and we say that X has a Bernoulli distribution. The expected value of X is

$$\mu = E(X) = (0)(1-p) + (1)(p) = p,$$

and the variance of X is

$$\sigma^2 = \text{Var}(X) = p^2(1-p) + (1-p)^2 p = p(1-p).$$

It follows that the standard deviation of X is $\sigma = \sqrt{p(1-p)}$.

In a sequence of n independent Bernoulli trials, where the probability of success remains constant, let X_i denote the Bernoulli random variable associated with the ith trial. In such a sequence of Bernoulli trials, we are often interested in the total number of successes and not in the order of their occurrence. If we let the random variable X equal the number of observed successes in n Bernoulli trials, the possible values of X are $0, 1, 2, \ldots, n$. If x successes occur, where $x = 0, 1, 2, \ldots, n$, then $n - x$ failures occur. The number of ways of selecting the x positions for the x successes in the n trials is

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

Since the trials are independent and the probabilities of success and failure on each trial are, respectively, p and $1-p$, the probability of each of these ways is $p^x(1-p)^{n-x}$. Thus the pmf of X is

$$p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 0, 1, 2, \ldots, n \\ 0 & \text{elsewhere.} \end{cases} \tag{17}$$

A random variable X that has a pmf of the form of $p(x)$ is said to have a binomial distribution, and any such $p(x)$ is called a binomial pmf. A binomial distribution is denoted by the symbol $b(n, p)$. The constants n and p are called the parameters of the binomial distribution.

Example 3.1. *Suppose we roll a fair six-sided die 3 times. What is the probability of getting exactly 2 sixes? For our notation, let X be the number of sixes obtained in the 3 rolls. Then X has a binomial distribution with $n = 3$ and $p = 1/6$. Hence,*

$$P(X = 2) = p(2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 0.06944.$$

The mgf of a binomial distribution is easily obtained as follows:

$$M(t) = \sum_x e^{tx} p(x) = \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x (1-p)^{n-x} = [(1-p) + pe^t]^n.$$

The mean μ and the variance σ^2 of X are

$$\mu = np, \quad \sigma^2 = np(1-p).$$

Theorem 3.1. *Suppose $X_1, X_2, \ldots, X_m$ are independent random variables such that X_i has binomial $b(n_i, p)$ distribution, for $i = 1, 2, \ldots, m$. Let $Y = \sum_{i=1}^m X_i$. Then Y has a binomial $b(\sum_{i=1}^m n_i, p)$ distribution.*

This result follows from the mgf or by convolution of pmfs.

1.1. Negative Binomial and Geometric Distributions. Consider a sequence of independent Bernoulli trials with constant probability p of success. Let the random variable Y denote the total number of failures in this sequence before the rth success, that is, $Y + r$ is equal to the number of trials necessary to produce exactly r successes with the last trial as a success. Here r is a fixed positive integer. The pmf of Y is

$$p_Y(y) = \begin{cases} \binom{y+r-1}{r-1} p^r (1-p)^y & y = 0, 1, 2, \ldots \\ 0 & \text{elsewhere.} \end{cases} \tag{18}$$

A distribution with a pmf of the form $p_Y(y)$ is called a negative binomial distribution. If $r = 1$, then Y has the pmf

$$p_Y(y) = p(1-p)^y, \quad y = 0, 1, 2, \ldots, \tag{19}$$

zero elsewhere. In this special case, $r = 1$, we say that Y has a geometric distribution. In terms of Bernoulli trials, Y is the number of failures until the first success.

The geometric distribution has the memoryless property:

$$P(Y \geq k + j \mid Y \geq k) = P(Y \geq j).$$

1.2. Multinomial Distribution. The binomial distribution is generalized to the multinomial distribution as follows. Let a random experiment be repeated n independent times. On each repetition, there is one and only one outcome from one of k categories. Call the categories $C_1, C_2, \ldots, C_k$. For each $i = 1, \ldots, k$, let p_i be the probability that the outcome is an element of C_i and assume that p_i remains constant throughout the n independent repetitions. Define the random variable X_i to be equal to the number of outcomes that are elements of C_i , $i = 1, 2, \ldots, k-1$. Because $X_k = n - X_1 - \cdots - X_{k-1}$, X_k is determined by the other X_i's.

The joint pmf of $(X_1, X_2, \ldots, X_{k-1})$ is

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_{k-1} = x_{k-1}) = \frac{n!}{x_1! \cdots x_{k-1}! x_k!} p_1^{x_1} \cdots p_{k-1}^{x_{k-1}} p_k^{x_k}, \tag{20}$$

for all $x_1, x_2, \ldots, x_{k-1}$ that are nonnegative integers and such that $x_1 + x_2 + \cdots + x_{k-1} \leq n$, where $x_k = n - x_1 - \cdots - x_{k-1}$ and $p_k = 1 - \sum_{j=1}^{k-1} p_j$.

We say that $(X_1, X_2, \ldots, X_{k-1})$ has a multinomial distribution with parameters n and $p_1, \ldots, p_{k-1}$.

1.3. Hypergeometric Distribution. Suppose we have a lot of N items of which D are defective. Let X denote the number of defective items in a sample of size n . If the sampling is done without replacement, the pmf of X is

$$p(x) = \frac{\binom{N-D}{n-x}\binom{D}{x}}{\binom{N}{n}}, \quad x = 0, 1, \ldots, n, \tag{21}$$

where, as usual, a binomial coefficient is taken to be 0 when the top value is less than the bottom value. We say that X has a hypergeometric distribution with parameters (N, D, n) .

The mean of X is

$$E(X) = n\frac{D}{N},$$

and the variance is

$$\mathrm{Var}(X) = n\frac{D}{N}\frac{N-D}{N}\frac{N-n}{N-1}. \tag{22}$$

The last term is often thought of as the correction term when sampling without replacement. Note that it is close to 1 if N is much larger than n .

2. The Poisson Distribution

Recall that the following series expansion holds for all real numbers z :

$$e^z = \sum_{x=0}^{\infty} \frac{z^x}{x!}.$$

Consider the function $p(x)$ defined by

$$p(x) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x = 0, 1, 2, \ldots \\ 0 & \text{elsewhere,} \end{cases} \tag{23}$$

where $\lambda > 0$. Since $\lambda > 0$, then $p(x) \geq 0$ and

$$\sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1;$$

that is, $p(x)$ satisfies the conditions of being a pmf of a discrete type of random variable. A random variable that has a pmf of the form $p(x)$ is said to have a Poisson distribution with parameter λ .

The Poisson distribution occurs in many areas of applications. Consider a process that counts the number of certain events occurring over an interval of time; for example, the number of tornados that touch down in Michigan per year, the number of cars entering a parking lot between 8:00 and 12:00 on a weekday, the number of car accidents at a busy intersection per week, the number of typographical errors per page of a manuscript, and the number of blemishes on a manufactured car door.

The mean and variance of a Poisson distribution are both λ :

$$E(X) = \lambda, \quad \mathrm{Var}(X) = \lambda.$$

The mgf of a Poisson distribution is

$$M(t) = \exp\{\lambda(e^t - 1)\}.$$

THEOREM 3.2. *If* $X_n \sim \mathrm{Bin}(n, p_n)$ *and* $np_n \to \lambda$ *as* $n \to \infty$ *, then* $X_n \xrightarrow{d} \mathrm{Poisson}(\lambda)$.

This theorem shows that the Poisson distribution can be used as an approximation to the binomial distribution when n is large and p is small.

3. The Γ, χ^2, and β Distributions

In this section we discuss several continuous distributions that are related to the gamma function.

3.1. The Gamma Distribution. The gamma function is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx, \quad \alpha > 0.$$

The gamma function satisfies the recursion $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$, and for positive integers, $\Gamma(n) = (n-1)!$.

Definition 3.1 (Gamma Distribution). *A random variable X is said to have a gamma distribution with parameters $\alpha > 0$ and $\beta > 0$ if its pdf is*

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & x > 0 \\ 0 & \textit{elsewhere.} \end{cases}$$

We write $X \sim \Gamma(\alpha, \beta)$.

The parameters α and β are called the shape and scale parameters, respectively. The mean and variance of X are

$$E(X) = \alpha\beta, \quad \mathrm{Var}(X) = \alpha\beta^2.$$

The mgf of a gamma distribution is

$$M(t) = (1-\beta t)^{-\alpha}, \quad t < \frac{1}{\beta}.$$

Two important special cases of the gamma distribution are:

- $\Gamma(1, \beta)$ is the exponential distribution $\mathrm{Exp}(1/\beta)$.
- $\Gamma(\alpha, 2)$ with $\alpha = n/2$ is related to the chi-square distribution.

If $X_1 \sim \Gamma(\alpha_1, \beta)$ and $X_2 \sim \Gamma(\alpha_2, \beta)$ are independent, then $X_1 + X_2 \sim \Gamma(\alpha_1 + \alpha_2, \beta)$.

3.2. The χ^2-Distribution.

Definition 3.2 (Chi-Square Distribution). *Let $Z_1, Z_2, \ldots, Z_n$ be iid $N(0,1)$ random variables. Then the random variable*

$$X = \sum_{i=1}^n Z_i^2$$

is said to have a chi-square distribution with n degrees of freedom. We write $X \sim \chi^2(n)$.

The chi-square distribution is a special case of the gamma distribution with $\alpha = n/2$ and $\beta = 2$. That is, $\chi^2(n) \equiv \Gamma(n/2, 2)$.

The pdf of $X \sim \chi^2(n)$ is

$$f(x) = \begin{cases} \frac{1}{\Gamma(n/2)2^{n/2}} x^{n/2-1} e^{-x/2} & x > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

The mean and variance of a chi-square distribution are

$$E(X) = n, \quad \mathrm{Var}(X) = 2n.$$

The chi-square distribution has the additive property: if $X_1 \sim \chi^2(m)$ and $X_2 \sim \chi^2(n)$ are independent, then $X_1 + X_2 \sim \chi^2(m+n)$.

3.3. The β-Distribution.

Definition 3.3 (Beta Distribution). *A random variable X is said to have a beta distribution with parameters $\alpha > 0$ and $\beta > 0$ if its pdf is*

$$f(x) = \begin{cases} \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1} & 0 < x < 1 \\ 0 & \textit{elsewhere,} \end{cases}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta function.

The mean and variance of $X \sim \mathrm{Beta}(\alpha, \beta)$ are

$$E(X) = \frac{\alpha}{\alpha+\beta}, \quad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Note that $\mathrm{Beta}(1, 1)$ is the uniform distribution on $(0, 1)$.

4. The Normal Distribution

The normal distribution is one of the most important distributions in statistics.

Definition 3.4 (Normal Distribution). *A random variable X is said to have a normal distribution with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ if its pdf is*

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$.

When $\mu = 0$ and $\sigma^2 = 1$, we say that X has a standard normal distribution, denoted by $Z \sim N(0, 1)$. The pdf of Z is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty,$$

and the cdf is $\Phi(z) = P(Z \leq z)$.

If $X \sim N(\mu, \sigma^2)$, then the standardized variable $Z = (X - \mu)/\sigma \sim N(0, 1)$. This property is extremely useful for computing probabilities:

$$P(X \leq x) = P\left(Z \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

The mean and variance of $X \sim N(\mu, \sigma^2)$ are $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$.

The mgf of X is

$$M(t) = \exp\left\{\mu t + \frac{\sigma^2 t^2}{2}\right\}.$$

THEOREM 3.3 (Linear Combinations of Normals). *If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a and b ,*

$$aX_1 + bX_2 \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2).$$

More generally, any linear combination of independent normal random variables is normally distributed.

5. The Multivariate Normal Distribution

5.1. Bivariate Normal Distribution. We say that (X_1, X_2) has a bivariate normal distribution if the joint pdf is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right\},$$

for $-\infty < x_1 < \infty$, $-\infty < x_2 < \infty$, where $\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $-1 < \rho < 1$. We write $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$.

The parameters have the following interpretations:

- $E(X_1) = \mu_1$ and $E(X_2) = \mu_2$.
- $\mathrm{Var}(X_1) = \sigma_1^2$ and $\mathrm{Var}(X_2) = \sigma_2^2$.
- ρ is the correlation coefficient between X_1 and X_2 .

THEOREM 3.4 (Properties of Bivariate Normal). *If $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, then:*

(1) The marginal distributions are $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$.

(2) The conditional distribution of X_2 given $X_1 = x_1$ is normal with mean and variance:

$$E(X_2 \mid x_1) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \quad \mathrm{Var}(X_2 \mid x_1) = \sigma_2^2(1-\rho^2).$$

(3) X_1 and X_2 are independent if and only if $\rho = 0$.

(4) Any linear combination $aX_1 + bX_2$ has a normal distribution.

6. t- and F-Distributions

6.1. The t-Distribution.

Definition 3.5 (t-Distribution). *Let $Z \sim N(0,1)$ and $X \sim \chi^2(n)$ be independent. Then the random variable*

$$T = \frac{Z}{\sqrt{X/n}}$$

is said to have a t-distribution with n degrees of freedom. We write $T \sim t(n)$.

The pdf of $T \sim t(n)$ is

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\,\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad -\infty < t < \infty.$$

The mean and variance of T are

$$E(T) = 0 \quad (n > 1), \quad \mathrm{Var}(T) = \frac{n}{n-2} \quad (n > 2).$$

As $n \to \infty$, $t(n) \xrightarrow{d} N(0,1)$.
When $n = 1$, the t-distribution is the Cauchy distribution.

6.2. The F-Distribution.

Definition 3.6 (F-Distribution). *Let $X_1 \sim \chi^2(m)$ and $X_2 \sim \chi^2(n)$ be independent. Then the random variable*

$$F = \frac{X_1/m}{X_2/n}$$

is said to have an F-distribution with degrees of freedom m and n . We write $F \sim F(m,n)$.

The pdf of $F \sim F(m,n)$ is

$$f(x) = \frac{\Gamma((m+n)/2)}{\Gamma(m/2)\Gamma(n/2)} \left(\frac{m}{n}\right)^{m/2} x^{m/2-1} \left(1 + \frac{mx}{n}\right)^{-(m+n)/2}, \quad x > 0.$$

The mean and variance of F are

$$E(F) = \frac{n}{n-2} \quad (n > 2), \quad \mathrm{Var}(F) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \quad (n > 4).$$

Note that if $F \sim F(m,n)$, then $1/F \sim F(n,m)$.

Exercises

EXERCISE 3.1. *3.1.1 If the mgf of a random variable* X *is* $\left(\frac{1}{3}+\frac{2}{3}e^t\right)^5$ *, find* $P(X = 2 \text{ or } 3)$ *.*

EXERCISE 3.2. *3.1.3 If* X *is* $b(n,p)$ *, show that* $E(X/n) = p$ *and* $E[(X/n-p)^2] = p(1-p)/n$ *.*

EXERCISE 3.3. *3.1.11 Toss two nickels and three dimes at random. Make appropriate assumptions and compute the probability that there are more heads showing on the nickels than on the dimes.*

EXERCISE 3.4. *3.2.1 Let* X *have a Poisson distribution with parameter* $\lambda = 3$ *. Find* $P(X = 2)$ *and* $P(X \geq 1)$ *.*

EXERCISE 3.5. *3.3.1 Show that the gamma function* $\Gamma(\alpha)$ *satisfies* $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ *.*

EXERCISE 3.6. *3.3.3 Let* $X \sim \chi^2(10)$ *. Find* $P(X > 15.99)$ *using R.*

EXERCISE 3.7. *3.4.1 Let* $X \sim N(100, 225)$ *(i.e.,* $\sigma = 15$ *). Find:*

(1) $P(X < 85)$ *;*
(2) $P(90 < X < 115)$ *;*
(3) $P(X > 120)$ *.*

EXERCISE 3.8. *3.4.3 If* $X_1 \sim N(50, 36)$ *and* $X_2 \sim N(30, 16)$ *are independent, find the distribution of* $Y = 2X_1 - 3X_2$ *.*

EXERCISE 3.9. *3.5.1 If* (X, Y) *has a bivariate normal distribution with* $\mu_1 = 2$ *,* $\mu_2 = 3$ *,* $\sigma_1^2 = 4$ *,* $\sigma_2^2 = 9$ *, and* $\rho = 1/2$ *, find* $E(Y \mid X = 2)$ *and* $\mathrm{Var}(Y \mid X = 2)$ *.*

EXERCISE 3.10. *3.6.1 Let* $T \sim t(10)$ *. Find* $P(|T| > 2.228)$ *.*

EXERCISE 3.11. *3.6.5 Let* $F \sim F(5, 10)$ *. Find* $E(F)$ *(if it exists) and* $P(F > 3.33)$ *.*

REMARK. ***Chapter 3 covers special distributions: binomial, negative binomial, Poisson, gamma, chi-square, beta, normal, multivariate normal,*** t***, and*** F ***distributions. Key formulas: (3.1.2) binomial pmf, (3.2.1) Poisson pmf, (3.3.1) gamma pdf, (3.4.1) normal pdf. Important relationships: binomial*** $\rightarrow$ ***Poisson (large*** n***, small*** p***), chi-square is special gamma,*** t ***and*** F ***involve normal and chi-square.***

CHAPTER 4

Some Elementary Statistical Inferences

1. Sampling and Statistics

In Chapter 2, we introduced the concepts of samples and statistics. We continue with this development in this chapter while introducing the main tools of inference: confidence intervals and tests of hypotheses.

In a typical statistical problem, we have a random variable X of interest, but its pdf $f(x)$ or pmf $p(x)$ is not known. Our ignorance about $f(x)$ or $p(x)$ can roughly be classified in one of two ways:

(1) $f(x)$ or $p(x)$ is completely unknown.

(2) The form of $f(x)$ or $p(x)$ is known down to a parameter θ , where θ may be a vector.

For now, we consider the second classification, although some of our discussion pertains to the first classification also. Some examples are the following:

(a) X has an exponential distribution $\text{Exp}(\theta)$, where θ is unknown.

(b) X has a binomial distribution $b(n, p)$, where n is known but p is unknown.

(c) X has a gamma distribution $\Gamma(\alpha, \beta)$, where both α and β are unknown.

(d) X has a normal distribution $N(\mu, \sigma^2)$, where both the mean μ and the variance σ^2 of X are unknown.

We often denote this problem by saying that the random variable X has a density or mass function of the form $f(x; \theta)$ or $p(x; \theta)$, where $\theta \in \Omega$ for a specified set Ω . For example, in (a) above, $\Omega = \{\theta \mid \theta > 0\}$. We call θ a parameter of the distribution. Because θ is unknown, we want to estimate it.

Suppose, in particular, we have a random variable X that has a distribution dependent on a parameter θ . Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from this distribution; that is, the X_i are iid (independent and identically distributed). A statistic is a function $T = g(X_1, X_2, \ldots, X_n)$ of the sample that does not depend on any unknown parameters. In this chapter we discuss several statistics that are used to estimate the unknown parameter θ or to test hypotheses about θ .

Let $X_1, X_2, \ldots, X_n$ denote a random sample of size n from a distribution that has mean $\mu = E(X_i)$ and variance $\sigma^2 = \text{Var}(X_i)$. The sample mean is defined by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{24}$$

The sample mean has the following properties:

$$E(\bar{X}) = \mu, \quad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

By the Weak Law of Large Numbers, $\bar{X}$ converges in probability to μ as $n \to \infty$. By the Central Limit Theorem, for large n ,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

The sample variance is defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2. \tag{25}$$

An unbiased estimator of σ^2 is S^2 because $E(S^2) = \sigma^2$.

Other important statistics include the sample standard deviation $S = \sqrt{S^2}$, the sample median, and the sample standard deviation.

1.1. Point Estimators. A point estimator $\hat{\theta}$ of a parameter θ is a statistic that provides a single value as an estimate of θ .

Several criteria are used to evaluate estimators:

Definition 4.1 (Bias). *The bias of an estimator $\hat{\theta}$ is defined as $E(\hat{\theta}) - \theta$. If $E(\hat{\theta}) = \theta$, then $\hat{\theta}$ is unbiased.*

Definition 4.2 (Mean Squared Error). *The mean squared error of an estimator $\hat{\theta}$ is* $\mathrm{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$. *For unbiased estimators,* $\mathrm{MSE} = \mathrm{Var}(\hat{\theta})$.

Example 4.1. *Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 . Then $\bar{X}$ is an unbiased estimator of μ because $E(\bar{X}) = \mu$. The variance of $\bar{X}$ is σ^2/n . The mean squared error of $\bar{X}$ is σ^2/n .*

Theorem 4.1 (Method of Moment Matching). *The kth sample moment is $M_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$. The method of moments equates sample moments to population moments to obtain estimators.*

Example 4.2. *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$. The maximum likelihood estimators (MLEs) are*

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Note that $\hat{\mu}$ is unbiased, but $\hat{\sigma}^2$ is biased (though asymptotically unbiased).

2. Confidence Intervals

A confidence interval provides a range of plausible values for an unknown parameter.

DEFINITION 4.3 (Confidence Interval). *Let θ be an unknown parameter. Let $L = g_1(X_1, \ldots, X_n)$ and $U = g_2(X_1, \ldots, X_n)$ be two statistics such that $L \leq U$. If for all $\theta \in \Omega$,*

$$P_\theta(L \leq \theta \leq U) = 1 - \alpha,$$

then the random interval $[L, U]$ is called a $100(1-\alpha)\%$ confidence interval for θ .

The quantity $1-\alpha$ is called the confidence coefficient, and α is the significance level.

EXAMPLE 4.3 (Confidence interval for normal mean, known variance). *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$ where σ^2 is known. Then a $100(1-\alpha)\%$ confidence interval for μ is*

$$\left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right],$$

where $z_{\alpha/2}$ satisfies $\Phi(z_{\alpha/2}) = 1 - \alpha/2$.

EXAMPLE 4.4 (Confidence interval for normal mean, unknown variance). *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown. Then a $100(1-\alpha)\%$ confidence interval for μ is*

$$\left[\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}, \quad \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right],$$

where $t_{\alpha/2,n-1}$ is the upper $\alpha/2$ critical value of the $t(n-1)$ distribution.

THEOREM 4.2 (Confidence interval for variance). *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$. A $100(1-\alpha)\%$ confidence interval for σ^2 is*

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \quad \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right],$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are critical values of the $\chi^2(n-1)$ distribution.

2.1. Confidence Intervals for Difference in Means. Let $X_1, \ldots, X_{n_1}$ be a random sample from $N(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_{n_2}$ be an independent random sample from $N(\mu_2, \sigma_2^2)$.

If σ_1^2 and σ_2^2 are known, a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\left[\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right].$$

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$ but unknown, the pooled estimator is

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}.$$

A $100(1-\alpha)\%$ confidence interval is

$$\left[\bar{X}-\bar{Y} \pm t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\right].$$

2.2. Confidence Interval for Difference in Proportions. Let $X \sim \operatorname{Bin}(n_1, p_1)$ and $Y \sim \operatorname{Bin}(n_2, p_2)$ be independent. Large-sample approximations yield the following $100(1-\alpha)\%$ confidence interval for $p_1 - p_2$:

$$\left[\hat{p}_1-\hat{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}\right],$$

where $\hat{p}_1 = X/n_1$ and $\hat{p}_2 = Y/n_2$.

3. Order Statistics

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with cdf $F(x)$. Define the order statistics by arranging the sample in ascending order:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}.$$

The kth order statistic is $X_{(k)}$. The sample median is $X_{((n+1)/2)}$ if n is odd, or the average of the two middle values if n is even.

The sample range is $R = X_{(n)} - X_{(1)}$ and the sample interquartile range is $IQR = X_{(3n/4)} - X_{(n/4)}$ (approximately).

The cdf and pdf of the kth order statistic $X_{(k)}$ are:

$$F_{X_{(k)}}(x) = \sum_{j=k}^{n} \binom{n}{j} [F(x)]^j [1-F(x)]^{n-j},$$

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1-F(x)]^{n-k} f(x).$$

3.1. Quantiles. The population quantile x_p is defined by $F(x_p) = p$.

Definition 4.4 (Sample Quantile). *The sample quantile x_p can be estimated by the order statistic $X_{(\lceil np \rceil)}$ or by linear interpolation.*

Common quantiles include:

- Median: $p = 0.5$
- Quartiles: $p = 0.25, 0.75$
- Deciles: $p = 0.1, 0.2, \ldots, 0.9$
- Percentiles: $p = 0.01, 0.02, \ldots, 0.99$

4. Introduction to Hypothesis Testing

A statistical hypothesis is a statement about the distribution of a random variable. We test a null hypothesis H_0 against an alternative hypothesis H_1 .

DEFINITION 4.5 (Null and Alternative Hypotheses). *The null hypothesis H_0 is the hypothesis being tested. The alternative hypothesis H_1 is the hypothesis we accept if H_0 is rejected.*

The four possible outcomes of a hypothesis test are:

	Fail to Reject H_0	Reject H_0
H_0 True	Correct	Type I Error (α)
H_0 False	Type II Error (β)	Correct Power ($1-\beta$)

The significance level $\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ true})$.

EXAMPLE 4.5 (Test for normal mean, known variance). *Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with known σ^2 . To test $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$, we use the test statistic*

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

We reject H_0 at level α if $Z > z_\alpha$.

EXAMPLE 4.6 (Test for normal mean, unknown variance). *Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with unknown σ^2 . To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we use the test statistic*

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

We reject H_0 at level α if $|T| > t_{\alpha/2,n-1}$.

5. Chi-Square Tests

The chi-square test is used for categorical data.

5.1. Goodness-of-Fit Test. Let $X_1, \ldots, X_n$ be iid observations from a categorical distribution with k categories. Under H_0, the probabilities are $p_{1,0}, p_{2,0}, \ldots, p_{k,0}$. The observed counts are $O_1, \ldots, O_k$ and the expected counts are $E_i = np_{i,0}$.

The chi-square goodness-of-fit statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

Under H_0 , this statistic has approximately a $\chi^2(k-1)$ distribution for large n . We reject H_0 if $\chi^2 > \chi^2_{\alpha,k-1}$.

5.2. Test of Independence. For a contingency table with r rows and c columns, let O_{ij} be the observed count in cell (i, j) . The expected count under independence is $E_{ij} = (O_{i\cdot}O_{\cdot j})/n$.

The chi-square test statistic for independence is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Under H_0 of independence, $\chi^2 \approx \chi^2((r-1)(c-1))$.

6. The Method of Monte Carlo

Monte Carlo methods use random sampling to approximate statistical quantities.

The law of large numbers states that if $E[g(X)] = \mu$ exists, then

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i) \xrightarrow{P} \mu \quad \text{as } n \to \infty.$$

Thus, we can approximate μ by simulating n values from the distribution of X and averaging $g(X_i)$.

Example 4.7 (Approximating π). *We can approximate π by simulating points in the unit square $[0, 1] \times [0, 1]$ and computing the proportion that fall inside the unit quarter circle. The area of the quarter circle is $\pi/4$, so $\hat{\pi} = 4 \times$ (proportion inside) .*

6.1. Accept-Reject Generation Algorithm. The accept-reject algorithm generates random samples from a target distribution $f(x)$ using a proposal distribution $g(x)$ with known density.

Example 4.8 (Accept-Reject Algorithm). *(1) Choose a proposal density $g(x)$ such that $f(x) \le Mg(x)$ for all x and some constant $M > 1$.*

(2) Generate $Y \sim g$ and $U \sim$ Uniform$(0, 1)$ independently.

(3) If $U \le f(Y)/(Mg(Y))$, accept $X = Y$; otherwise, reject and return to step 2.

The acceptance probability is $1/M$. The algorithm generates samples from f exactly.

7. Bootstrap Procedures

Bootstrap methods use resampling to estimate the sampling distribution of a statistic.

Let $X_1, \ldots, X_n$ be a random sample and $T = g(X_1, \ldots, X_n)$ be a statistic. The bootstrap proceeds as follows:

(1) Resample n observations with replacement from the original sample to form a bootstrap sample $X_1^*, \ldots, X_n^*$.

(2) Compute the bootstrap replicate $T^* = g(X_1^*, \ldots, X_n^*)$.

(3) Repeat steps 1-2 many times (say B times) to obtain $T_1^*, \ldots, T_B^*$.
(4) Approximate the distribution of T by the empirical distribution of $T_1^*, \ldots, T_B^*$.

7.1. Percentile Bootstrap Confidence Intervals. A $100(1-\alpha)\%$ percentile bootstrap confidence interval for θ is

$$\left[T^*_{(\alpha/2)}, T^*_{(1-\alpha/2)}\right],$$

where $T^*_{(k)}$ is the kth order statistic of the bootstrap replicates.

Exercises

Exercise 4.1. *4.1.1 Let X_1, X_2, X_3, X_4, X_5 be a random sample of size 5 from a distribution with mean $\mu = 10$ and variance $\sigma^2 = 4$. Find the mean and variance of $\bar{X}$.*

Exercise 4.2. *4.1.3 Let $X_1, \ldots, X_n$ be iid with mean μ and variance σ^2 . Show that S^2 is an unbiased estimator of σ^2 .*

Exercise 4.3. *4.2.1 Let $X_1, \ldots, X_{16}$ be a random sample from $N(50, 25)$. Find $P(48 < \bar{X} < 52)$.*

Exercise 4.4. *4.2.3 Construct a 95% confidence interval for the mean of a normal distribution based on the following sample: $n = 25$, $\bar{x} = 100$, $s = 15$.*

Exercise 4.5. *4.3.1 Two independent samples are given: Sample 1: $n_1 = 30$, $\bar{x}_1 = 75$, $s_1 = 10$; Sample 2: $n_2 = 40$, $\bar{x}_2 = 80$, $s_2 = 12$. Assuming equal variances, find a 90% confidence interval for $\mu_1 - \mu_2$.*

Exercise 4.6. *4.4.1 Let $X_1, \ldots, X_{10}$ be a random sample from $N(0, 1)$. Find the probability that the sample median exceeds 0.5.*

Exercise 4.7. *4.5.1 A manufacturer claims that the mean lifetime of a light bulb is at least 1000 hours. A sample of 25 bulbs has a mean lifetime of 950 hours with a standard deviation of 80 hours. Test the claim at the $\alpha = 0.05$ level.*

Exercise 4.8. *4.7.1 A chi-square goodness-of-fit test is performed on a die to test whether it is fair. The die is rolled 120 times with the following frequencies: Face 1: 18, Face 2: 22, Face 3: 16, Face 4: 24, Face 5: 19, Face 6: 21. Compute the test statistic and p-value.*

Exercise 4.9. *4.8.1 Describe the accept-reject algorithm for generating random samples from a gamma distribution using an exponential proposal distribution.*

Exercise 4.10. *4.9.1 Explain the bootstrap procedure for estimating the standard error of the sample median.*

Remark. ***Chapter 4 covers statistical inference: sampling, point estimation, confidence intervals, hypothesis testing, chi-square tests, Monte Carlo methods, and bootstrap procedures. Key concepts: sample mean and variance, unbiased estimation, CI formulas for normal means and variances, hypothesis testing framework, Type I/II errors, bootstrap resampling.***

1. 问答记录

问题记录. 问题: ... **回答:** ...

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